10TH CLASS BIOLOGY — NEW SYLLABUS (PECTAA)

Chapter 10 — Evolution

Solved Exercise | New Book (PECTAA)

Exercise (Solved)

A. Select the correct answers for the following questions.

1. The survival and reproduction of individuals with advantageous traits is called:

(a) Mutations (b) Gene flow
(c) Natural selection (d) Asexual reproduction

Answer: (c) — Natural selection is the process through which the individuals with better characteristics (variations) are more likely to live, reproduce and pass those variations to their offspring. In this way the favourable variations become more common in the population.

2. A population is split by a river. After many generations, both populations evolve into different species. What type of speciation may occur?

(a) Sympatric (b) Genetic drift
(c) Mutation (d) Allopatric

Answer: (d) — Allopatric speciation occurs when a population is geographically separated into two or more populations by a physical barrier such as a mountain range, a river or an ocean. The two groups cannot breed with each other, so they evolve differently and become separate species.

3. Which characteristic in whales supports that their ancestors lived on land?

(a) Big body (b) Fins
(c) Pelvic bones (d) Stream-lined body

Answer: (c) — Modern whales have small, non-functional pelvic bones even though they have no legs. These are vestigial structures which show that the ancestors of whales had legs and lived on land.

4. Farmers choose best cows to breed. It is an example of:

(a) Natural selection (b) Selective breeding
(c) Adaptation (d) Genetic mutation

Answer: (b) — Selective breeding is the process in which humans select varieties of plants or animals and breed them to produce offspring with the required characteristics, for example to increase the production of milk. Here the selection is done by humans, not by nature.

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5. Darwin's observation of finches proved that:

(a) Climate was same on all islands
(b) All finches were identical
(c) They adapted to different foods on different islands
(d) Finches never migrated

Answer: (c) — On the Galapagos Islands Darwin noticed that different species of finches had different sizes and shapes of beaks on different islands, because different foods such as seeds, cacti and insects were available there. Their beaks were suited to the food available on each island, so they had adapted to different foods on different islands.

6. A new variation appears suddenly and helps survival. What happens next?

(a) It disappears
(b) It spreads through the population
(c) It causes disease
(d) It is removed by natural selection

Answer: (b) — A variation which helps survival is a favourable variation, so the individuals having it live longer and reproduce more. They pass this variation to their offspring, and over many generations it becomes more common in the population, that is, it spreads through the population.

7. Which of these is a vestigial structure?

(a) Human ear muscles
(b) Wings of sparrow
(c) Beak of duck
(d) Tail of cat

Answer: (a) — Vestigial organs are body parts which have no apparent function now but were fully functional in the ancestors. In animals such as cats and dogs the ear muscles move the ears to detect sounds, but in humans these ear muscles are largely non-functional, so they are vestigial structures.

8. Which pair are homologous structures?

(a) Bird wings and butterfly wings
(b) Dolphin fins and shark fins
(c) Human arms and dolphin fins
(d) Bat wings and bird wings

Answer: (c) — Homologous structures are similar in structure but perform different functions, and they show evolution from a common ancestor. The human arm and the dolphin fin have the same basic structure of bones but different functions. Bird and butterfly wings are only functionally similar, so they are analogous organs.

9. Human appendix is a vestigial organ because it:

(a) Has a major digestive function
(b) Is part of the immune system
(c) Is no longer useful for digestion
(d) Helps in absorbing nutrients

Answer: (c) — The appendix is a small, tube-like structure attached to the caecum. It might have played a more important role in the digestive system of our ancestors, but in humans it is no longer useful for digestion, so it is a vestigial structure.

10. Artificial selection is similar to natural selection because:

(a) Environmental conditions play role in both
(b) Both occur over millions of years
(c) Both are carried out by humans
(d) Both depend upon variations among individuals

Answer: (d) — In both natural and artificial (selective) selection the starting material is the naturally occurring variations among the individuals of a species. In artificial selection the human being chooses the useful variations, while in natural selection the environment favours them; the selection itself is not carried out by humans in nature.

B. Write short answers.

1. Write a short note on selective breeding.

Selective breeding is the process in which humans select varieties of plants or animals and breed them to produce offspring with the required characteristics. In this way characteristics such as grain production and disease resistance can be improved in plants, and growth rate or milk production can be improved in animals. The varieties of animals which are selectively bred are called breeds, and the varieties of plants are called varieties or cultivars. Many breeds of sheep, goat, cow and hen have been produced to increase the production of meat, milk, eggs and wool, and many plant cultivars of wheat, rice, potato and apple have been produced for better quantity and quality. Selective breeding thus provides evidence of the naturally occurring variations among species and their evolution.

2. What are the two types of speciation?

(i) Allopatric speciation: it occurs when a population is geographically separated into two or more populations. The physical barrier, such as a mountain range, a river or an ocean, prevents the individuals of the different groups from breeding. Over time the isolated populations evolve differently and become distinct species; for example, the finches of the different Galapagos Islands became separate species.

(ii) Sympatric speciation: it occurs when reproductive isolation happens within the same geographical area because of habitat preference or behaviour. For example, groups of fishes of the same species live in the same lake but are reproductively isolated because of their dietary preferences or mate selection, and over time these groups become new species.

3. Provide examples of homologous structures.

Homologous structures are the similar structures present in different species which perform different functions. The best example is the forelimbs of mammals: the arm of a human, the leg of a cat, the flipper of a whale and the wing of a bat all have the same basic structure of bones, but they are used for different functions such as grasping, walking, swimming and flying. This basic similarity of structure is the evidence that these animals evolved from a common ancestor. Their structural similarity shows relationship, while their different functions show adaptation to different ways of life.

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4. List the steps in the process of natural selection.

Natural selection works in the following steps:

Step	Explanation with example
1. Variations	The individuals of a species are slightly different from one another in colour, size, speed, strength or ability to find food. For example, some beetles are green and some are brown.
2. Struggle for survival	Living things compete for limited resources such as food, water and space, and they also face predators, diseases and harsh environments. For example, birds eat beetles, so beetles must survive by hiding or blending in.
3. Survival of the fittest	The individuals with better variations are more likely to live and reproduce. “Fittest” means best suited, not necessarily the strongest; brown beetles blend in with the soil, so birds hunt them less.
4. Reproduction and passing on variations	The surviving individuals reproduce and pass their useful variations to their offspring, so these variations become more common; more brown beetles are born.
5. Gradual evolution over time	After many generations the small changes build up and the species slowly changes. If the environment keeps favouring certain variations, new species may form, and the whole beetle population may become brown.

C. Write answers in detail.

1. Explain the theory of evolution by natural selection.

Evolution is the change in the heritable traits of a population over the course of successive generations. In the 1850s Charles Darwin explained the mechanism of evolution and published it in 1859 in his book “On the Origin of Species”.

The theory of evolution by natural selection states that living things change over time. Those with helpful characteristics (variations) survive, reproduce and pass those characteristics to their young. Over many generations these favourable characteristics become more common in the population, and this leads to the evolution of new species.

The process works in five steps. First, variations exist among the individuals of every species. Second, there is a struggle for survival, because the resources are limited and the organisms face predators, diseases and harsh environments. Third, the individuals with better variations survive, that is, the survival of the fittest, where “fittest” means best suited and not necessarily the strongest. Fourth, the surviving

individuals reproduce and pass the useful variations to their offspring, so these variations become more common. Fifth, after many generations the small changes accumulate and a new species may be formed.

A good example is the giraffe's long neck. Some giraffes had slightly longer necks than others, and the leaves on tall trees were hard to reach. The long-necked giraffes could eat leaves from taller trees and survived better in harsh conditions, so they reproduced and passed the long-neck trait to their babies, and after many generations most giraffes had long necks. Another example is the peppered moth: after pollution darkened the tree bark during the industrial revolution in England, the dark-coloured moths blended in and survived better than the light ones, so the dark moths reproduced more and the population gradually became dark.

2. Write a note on the observations Darwin made during his voyage.

Charles Darwin (1809–1882) was an English naturalist who went on a five-year trip around the world on His Majesty's Ship (HMS) Beagle. During this voyage he made many important observations which laid the foundations of his theory of evolution by natural selection.

(i) Unique species: he observed a variety of unique plant and animal species in different regions. He noticed that different islands and environments had their own distinct sets of species.

(ii) Variations in the same species: he observed variations among the same species living in different environments. These were adaptations, that is, special variations in an organism's body or behaviour which help it to survive in its environment, and he was convinced of the role of adaptations in the survival of species.

(iii) Beaks of finches: on the Galapagos Islands he noticed that different species of finches had different sizes and shapes of beaks on different islands, because different foods such as seeds, cacti and insects were available there. The beaks of the finches were suited to the food sources available on each island.

(iv) Fossils of extinct animals: when he observed the fossils of extinct animals, he found their resemblance with modern species. This suggested that species could change over time.

After returning from the voyage Darwin analysed these observations, gathered more evidence and published his work in 1859.

3. Describe the sources of variation.

Variations are the inheritable differences among the individuals of a population, and natural selection can act only on these variations. Their main sources are the following.

(i) Mutation: it is a permanent change in the DNA of an organism which can pass to the offspring. Mutations can occur spontaneously or due to environmental factors such as radiations or chemicals, and they produce new characteristics. Most mutations are neutral or harmful, but some are beneficial; for example, a mutation may produce antibiotic resistance in some bacteria, and these resistant bacteria then survive and reproduce in the presence of antibiotics.

(ii) Genetic recombination: in sexually reproducing organisms, genetic recombination occurs during the formation of gametes through meiosis. The mixing of the genetic material of two parents produces offspring with new combinations of genes, which is why human siblings have different combinations of characteristics such as hair colour and eye colour.

(iii) Gene flow (migration): it occurs when individuals from one population migrate to another population and introduce new genes in it. If these new variations prove better in the new population, they spread through it; for example, plants of one area cross-pollinating with plants of another region introduce new characteristics such as improved drought resistance.

(iv) Random pairing of gametes: during sexual reproduction the fusion of the egg and the sperm is random, because any sperm can fertilize any egg. This produces offspring with unique combinations of genes, which increases variation; for example, the children of the same parents are different from one another.

4. Justify how fossil record provides an evidence of evolution.

Fossils are the remains, impressions or traces of organisms that lived in the past, and most fossils are found in sedimentary rocks. They provide a visual record of evolution, because the deeper the layer of the rock, the older the fossil present in it; so the fossils of different layers show a sequence of change.

Palaeontologists study fossils to observe the sequence of evolutionary changes among organisms. For example, the fossils of the vertebrate classes show that fishes are the earliest vertebrates, followed by amphibians, reptiles, and then birds and mammals. The fossils of organisms also show the characteristics of the ancestors together with the new features of their descendants, so they bridge the gaps in the evolutionary history of various species and provide evidence of gradual change over time.

A very good example is the fossil of Archaeopteryx, a bird-like dinosaur. It has features of both reptiles and birds, and it provides evidence that the birds evolved from a tetrapod dinosaur, that is, from a reptile. Such intermediate fossils cannot be explained without evolution, so the fossil record is a strong evidence of evolution.

5. Write a note on homologous structures as an evidence of evolution.

Comparative anatomy involves the study of the similarities and differences in the structures of different organisms. The similar structures present in different species which perform different functions are called homologous structures.

The best example is the forelimbs of mammals: the arm of a human, the leg of a cat, the flipper of a whale and the wing of a bat all have the same basic structure and the same arrangement of bones, but they perform very different functions, such as grasping, walking, swimming and flying. The basic similarity of structure of these forelimbs is the evidence that these animals evolved from a common ancestor, while their different functions show that each of them became adapted to a different way of life.

Homologous structures must not be confused with analogous structures. Analogous organs are functionally the same but structurally different, for example the wings of bat, birds and insects; they show similar functions but do not show a common

ancestry. Thus homologous structures show relationship by descent, which is why they are an important evidence of evolution.

6. What are vestigial structures? Explain how they provide evidence of evolution.

Vestigial organs are body parts which have no apparent function now, but which were fully functional in the ancestors of the organism. It is believed that through evolution these organs lost their original function over time, because the organism stopped using them.

Examples: the human appendix, a small tube-like structure attached to the caecum, which might have played a more important role in the digestive system of our ancestors; the ear muscles of humans, which are largely non-functional although in cats and dogs they move the ears to detect sounds from different directions; the pelvic bones of whales, which are small and non-functional although whales have no legs; and the wings of flightless birds such as the ostrich, which are not used for flight.

These structures provide evidence of evolution because a useless organ would not be present in a body which was designed perfectly for its present way of life. Their presence shows that the present organism has descended from an ancestor in which these organs were useful, and that the organ gradually reduced when the way of life changed. For example, the small pelvic bones inside a whale tell us that its ancestors had legs and lived on land. In this way vestigial structures are like a record of the evolutionary past written inside the body.

D. Inquisitive questions

1. Why is variation among individuals important for evolution?

Variation is the raw material of evolution, because natural selection can act only on the differences which already exist among the individuals of a population. If all the individuals were exactly the same, then all of them would face a change in the environment in the same way: either all would survive or all would die, and no species could improve. But because the individuals differ in size, colour, speed and strength, some of them are better suited to the new conditions. These individuals survive and reproduce, while the others do not, and the useful variations are passed to the next generation. Over many generations these favourable variations become common in the population, and if the changes continue, a new species may be formed. So without variation there would be no natural selection and no evolution.

2. How do fossils help scientists understand the history of life on Earth?

Fossils are the remains, impressions or traces of organisms that lived in the past, and most of them are found in sedimentary rocks. As the layers of rock are formed one above the other, the deeper layers are older, so the position of a fossil tells the scientist its age and its place in the history of life.

By studying fossils scientists can see which kinds of organisms lived in different periods, and in which order they appeared. For example, the fossils of vertebrates show that fishes appeared first, then amphibians, reptiles, and finally birds and mammals. The fossils also show the characteristics of the ancestors together with

the new features of their descendants, so they bridge the gaps in the evolutionary history of many species. Intermediate fossils such as Archaeopteryx show how birds could have evolved from reptiles, and the fossils of extinct animals which resemble the modern species show that species change over time. In this way the fossil record gives a visual proof of evolution and tells us the story of life on Earth.

3. Why can isolation lead to the formation of new species?

When a population is separated by a physical barrier such as a river, a mountain range or an ocean, the two groups can no longer breed with each other, so no genes are exchanged between them. Each group now faces slightly different conditions of food, climate and predators, and different variations are useful in each of them. Natural selection therefore works in a different direction in the two groups, and the favourable variations which spread in one group are not the same as in the other. Moreover, mutations and genetic recombination keep adding new differences in both groups separately.

After many generations the two groups become so different in their body structure and behaviour that they can no longer interbreed and produce fertile offspring under natural conditions. At that stage they are two different species; this is called allopatric speciation. Isolation can also happen within the same area because of differences in habitat or mating behaviour, which leads to sympatric speciation. Thus isolation stops the mixing of genes and allows the separated populations to evolve independently into new species.

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